COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	8/28/26	**DATE SUBMITTED:**	9/2/26
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ACTIVITY NO. 6.1 Searching Techniques

1. PROCEDURE OUTPUT

What procedures did you perform?

```cpp
#include <iostream>
#include <cstdlib> //for generating random integers
#include <time.h>  //will be used for our searching function
#include "searchingAlgo.h"

const int max_Size = 50;

int main()
{
    /*
        // generate random values:
         int dataset[max_size];
         srand(time(0));
         for(int i = 0; i < max_size; i++){
             dataset[i]=rand();
             std::cout<< dataset[i] << " ";
         }
         std::cout<<std::endl;

         int choice;
         std::cout <<"Indicate the numbser to search: ";
         std::cin>> choice;

         linearSearchArr(max_size, choice, dataset);

         Node<char>* name1 = new_node('Y');
         Node<char>* name2 = new_node('O');
         Node<char>* name3 = new_node('O');
         Node<char>* name4 = new_node('O');
         Node<char>* name5 = new_node('O');
         Node<char>* name6 = new_node('O');
```

```
32          Node<char>* name7 = new_node('O');
33
34          name1->next = name2;
35          name2->next = name3;
36          name3->next = name4;
37          name4->next = name5;
38          name5->next = name6;
39          name6->next = name7;
40
41          linearSearchLL(name1, 'D');
42          linearSearchLL(name1, 'X');
43
44          // generate a sorted list for the array:
45          int dataset2[max_size];
46          for(int i = 0; i < max_size; i++){
47              dataset2[i] = i;
48              std::cout << dataset2[i] << " ";
49          }
50
51          binarySearchArr(11, dataset2, max_size);
52          binarySearchArr(11111, dataset2, max_size);
53      }
54      */
55      int dataset2[max_Size];
56      for (int i = 0; i < max_Size; i++)
57      {
58          dataset2[i] = i;
59          std::cout << dataset2[i];
60      }
```

```
54      */
55      int dataset2[max_Size];
56      for (int i = 0; i < max_Size; i++)
57      {
58          dataset2[i] = i;
59          std::cout << dataset2[i];
60      }
61      return 0;
62  }
```

HOA6.1 > C node.h > new_node<T>(T)

```
1   #ifndef NODE_H
2   #define NODE_H
3
4   template <typename T>
5   class Node
6   {
7   public:
8       T data;
9       Node *next;
10  };
11
12  template <typename T>
13  Node<T> *new_node(T newData)
14  {
15      Node<T> *newNode = new Node<T>;
16      newNode->data = newData;
17      newNode->next = nullptr;
18      return newNode;
19  }
20
21  #endif
22
```

```cpp
#include "node.h"

#ifndef SEARCHINGALGO_H
#define SEARCHINGALGO_H

template <typename T>
void linearSearchArr(int maxSize, T dataFind, T data[])
{
    int i = 0;
    while (i <= maxSize)
    {
        if (dataFind == data[i])
        {
            std::cout << "Searching is successful\n";
            return;
        }
        i++;
    }
    std::cout << "the search is unsuccessful\n";
}

template <typename T>
void linearSearchLL(Node<T> *node, T dataFind)
{

    // CREATE A LOOP THAT ITERATES UNTIL THE NODE REACHES THE END (NULLPTR)
    while (node != nullptr)
    {
        // CREATE AN IF CONDITION THAT WILL COMPARE THE SEARCH ITEM TO THE CURRENT NODE DATA
        if (node->data == dataFind)
        {
```

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```
32              std::cout << "the search was success.\n";
33              return;
34          }
35          // ITERATOR OR / MOVING OF NDOE
36          node = node->next;
37      }
38      // DISPLAY IF UNSUCCESSFUL
39
40      std::cout << "the search was unsuccessful.\n";
41  }
42
43  template <typename T>
44  void binarySearchArr(T dataFind, T data[], int max_Size)
45  {
46      int low = 0, up = maxSize - 1;
47      while (low <= up)
48      {
49          middle = (int)((low + up) / 2);
50          if (dataFind == data[middle])
51          {
52              std::cout << "searching is success.\n";
53          }
54          else if (dataFind > data(middle))
55          {
56              low = middle + 1;
57          }
58          else
```

```
58          else
59          {
60              up = middle - 1;
61          }
62      }
63      std::cout << "search element was not found.";
64  }
65
66  // create a helper function to get the middle code
67  template <typename T>
68  Node<T> *getMiddle(Node<T> *start, Node<T> *last)
69  {
70      // use two pointers
71      // slow moves one node at a time
72      // fast moves two at a time
73
74      Node<T> *slow = start;
75      Node<T> *fast = start;
76
77      // create a loop until fast reaches the end
78      // or reaches the last node of the current search range
79
80      while (fast != last && fast->next != last)
81      {
82          slow = slow->next;
83          fast = fast->next;
84      }
```

```
84      }
85      return slow;
86  }
87  template <typename T>
88
89  void binarySearchLL(Node<T> *node, T dataFind)
90  {
91
92      Node<T> *last = nullptr;
93
94      while (start != last)
95      {
96          // middle element
97          middle = getMiddle(start, last);
98
99          if (middle->data == dataFind)
100         {
101             std::cout << "the data was found.\n";
102             return;
103         }
104         else if (middle->data > dataFind)
105         {
106             last = middle;
107         }
108         else
109         {
110             start = middle->next;s
```

```
110             start = middle->next;s
111         }
112     }
113     std::cout << "search element was not found.\n";
114 }
115 #endif
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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